

Exercise 0. Let H be a Hilbert space. Give a direct proof (without appeal to the Hahn-Banach theorem) of the following statement:
If H' is a subspace of H , then any bounded linear functional on H' extends to a bounded linear functional on H (actually with the same bound).

Exercise 1. Let

$$S = \{f_n : [-1, 1] \rightarrow \mathbb{R} : f_n(x) = x^{2n}\} \subset L^2([-1, 1]).$$

Determine the closure of $\text{span} S$ and $S^\perp$.

Exercise 2. Let $A \subset \mathbb{R}^n$, $B \subset \mathbb{R}^m$ be open sets and let $\{f_n\}$ be an orthonormal basis in $L^2(A)$, $\{g_n\}$ be an orthonormal basis in $L^2(B)$. Use these functions to construct an orthonormal basis in $L^2(A \times B)$.

Exercise 3. Show that if $\{x_n\}$ is an orthonormal basis in the Hilbert space H and $\{y_n\}$ is an orthonormal sequence satisfying

$$\sum_n \|x_n - y_n\|^2 < 1$$

then $\{y_n\}$ is an orthonormal basis in H .

Exercise 3. Let $\{f_n : n \geq 1\}$ be an orthonormal system of real-valued functions in $L^2([0, 1])$ such that

$$\sum_{n \geq 1} \left(\int_0^x f_n(t) dt \right)^2 = x, \quad x \in [0, 1].$$

Show that $\{f_n : n \geq 1\}$ is an orthonormal basis.

Exercise 4. Let $H_1, H_2, \dots, H_n, \dots$ be a countable family of Hilbert spaces. Let H be the set of tuples $x = (x_1, \dots, x_n, \dots) \in \prod_n H_n$ with the property that

$$\|x\|^2 = \sum_n \|x_n\|_{H_n}^2 < \infty.$$

Show that with respect to $\|\cdot\|$, H is a Hilbert space.

Exercise 5. Let H be a Hilbert space and let Y, Z be two closed subspaces of H . Let $\alpha(Y, Z) \in [0, \pi/2]$ be uniquely determined by

$$\cos \alpha(Y, Z) = \sup\{|\langle y, z \rangle| : y \in Y, z \in Z, \|y\| = \|z\| = 1\}.$$

This number is called the angle between Y and Z . Show that:

- a) $Z \subset Y^\perp$ if and only if $\alpha(Y, Z) = \pi/2$,
- b) $\alpha(Y, Z) > 0$ if and only if for every $\epsilon > 0$ there exists $\delta > 0$ such that whenever $y \in Y, z \in Z$ satisfy $\|y - z\| < \delta$, then $\|y\|, \|z\| < \epsilon$.

Exercise 6. Show that if H is finite dimensional $\alpha(Y, Z) > 0$ if and only if $Y \cap Z = \{0\}$.

Exercise 7. Construct a Hilbert space H and two closed subspaces Y, Z with $Y \cap Z = \{0\}$ and $\alpha(Y, Z) = 0$.